
DESCRIPTIVE COMPANION NOTE

Inflation Volatility, Essential Price Dispersion, and Household Planning Risk

A Public-Data Companion Note on Budget Instability and Must-Pay Prices

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DOCUMENT TYPE

Descriptive companion note

DATE

May 2026

Abstract

Headline CPI is an incomplete description of household budget instability, but aggregate public price series cannot by themselves measure household “planning damage.” This revised companion note narrows the claim. It studies selected essential CPI categories—food at home, rent of primary residence, electricity, gasoline, and medical care—to show how must-pay prices can move unevenly, at different times, and with different price-level persistence. The note constructs transparent descriptive measures: an equal-weighted selected-essential inflation mean, a median alternative, an ex-gasoline mean, cross-category dispersion, ex-gasoline dispersion, rolling 12-month volatility, and cumulative price-level indexes from a December 2019 baseline. The strongest finding is also a caution: the December 2021 selected-essential mean of 13.7 percent and dispersion of 20.5 percentage points were heavily influenced by gasoline. Excluding gasoline, the selected-essential mean was 4.6 percent and dispersion was 2.2 percentage points. The useful contribution is therefore not a validated household-stress index. It is a disciplined measurement warning: household planning risk should distinguish inflation levels, cumulative price levels, cross-category dispersion, volatility, exposure, and realized hardship. A publishable empirical paper would need household-specific expenditure shares and outcomes such as missed payments, borrowing, delayed care, savings depletion, or subjective financial stress.

1. Motivation

A household does not pay headline CPI. It pays rent or mortgage-related costs, food, utilities, transportation, insurance, medical bills, debt service, and other obligations on different schedules. Headline CPI is useful for macroeconomic comparison, but household planning depends on cash-flow categories.

The earlier version of this note used the phrase “planning damage.” The external review was right: that phrase is too strong unless damage is measured directly. This revision uses **household planning risk** and **budget instability** instead. The note measures selected price inputs that may contribute to planning risk. It does not measure hardship, forecast error, or welfare loss.

The practical question is still worth asking. When must-pay categories move unevenly or at staggered times, households may not experience immediate relief when headline inflation cools. But that statement must be kept separate from the stronger causal claim that price dispersion causes hardship.

2. Definitions and formulas

Let $\pi_{i,ct}$ be the year-over-year percent change in CPI category c at month t . The selected essential categories are food at home, rent of primary residence, electricity, gasoline, and medical care. This is a selected CPI basket, not a full necessities index and not a household-specific cost-of-living measure.

The equal-weighted selected-essential mean is:

$$E_t = (1/C) \sum_c \pi_{ct}.$$

The cross-category dispersion measure is the sample standard deviation:

$$D_t = \sqrt{(1/(C-1)) \sum_c (\pi_{ct} - E_t)^2}.$$

The rolling volatility measure is:

$$V_t = \text{sd}(E_t, E_{t-1}, \dots, E_{t-11}).$$

The note also reports a median inflation alternative, an ex-gasoline mean, an ex-gasoline dispersion measure, and a median absolute deviation robustness check. CPI inflation rates are simple percent changes, not log differences. The annual table reports December year-over-year CPI measures. Sentiment is annual average and is included only as context; it is not modeled as an outcome.

The public-data panel is complete through December 2024 in the fetched FRED component files used here. Later partial observations are excluded from the annual tables and figures to avoid mixing complete and incomplete component coverage.

3. What this note can and cannot measure

The note distinguishes six concepts that are easy to blur:

1. **Level pressure:** prices are high relative to income.
2. **Inflation pressure:** prices are rising quickly.
3. **Cross-category dispersion:** selected categories are rising at different rates.
4. **Realized volatility:** a measured inflation series fluctuates over time.
5. **Uncertainty:** households cannot predict future bills.
6. **Welfare damage:** households experience costly adjustment or hardship.

This note measures items 2 through 4 and adds a price-level index for item 1. It does not measure uncertainty or welfare damage. Household-specific planning risk would require exposure weights and outcomes.

4. Selected essential inflation and robustness

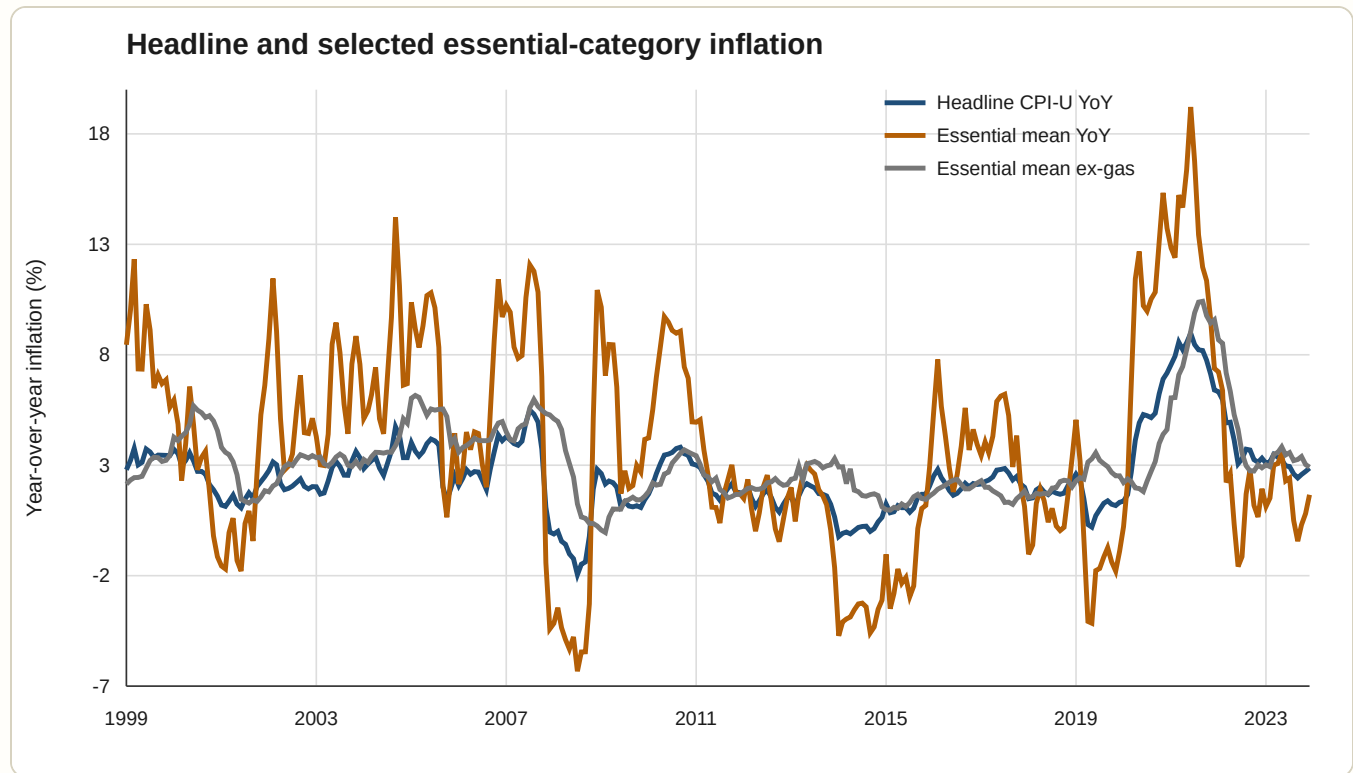


Figure 1. Headline CPI-U inflation, selected-essential mean inflation, and selected-essential mean excluding gasoline.

Figure 1 compares headline CPI-U with the equal-weighted selected-essential mean and an ex-gasoline version. The ex-gasoline line matters because gasoline can dominate a five-category equal-weighted basket. The figure should therefore be read as a measurement experiment, not as proof that the selected-essential mean is superior to headline CPI.

Table 1. Recent public-data planning-risk indicators.

Year	Headline YoY	Essential mean YoY	Essential median YoY	Essential mean ex-gas	Dispersion SD	Dispersion SD ex-gas	Sentiment avg
2019	2.3	3.3	3.7	2.0	3.6	2.5	96.0
2020	1.3	-0.9	2.0	2.5	7.7	1.0	81.5
2021	7.2	13.7	6.4	4.6	20.5	2.2	77.6
2022	6.4	7.4	8.3	9.6	6.3	4.5	59.0
2023	3.3	1.9	1.3	2.9	3.1	2.7	65.3
2024	2.9	1.7	2.8	2.9	2.9	1.0	72.5

Source note: FRED-hosted CPI component series and University of Michigan consumer sentiment. CPI entries are December year-over-year percent changes except sentiment, which is the annual average of monthly readings and included only as broad context. Dispersion is in percentage points and is calculated as the sample standard deviation across selected essential categories.

The December 2021 values show why robustness is necessary. The selected-essential mean was 13.7 percent and dispersion was 20.5 percentage points, but that signal was heavily affected by gasoline.

Table 2. December 2021 robustness checks.

Measure	December 2021 value
Essential mean YoY	13.7
Essential median YoY	6.4
Essential mean YoY excluding gasoline	4.6
Dispersion SD	20.5
Median absolute deviation	3.1
Dispersion SD excluding gasoline	2.2

The robustness table changes the interpretation. The December 2021 essential mean excluding gasoline was 4.6 percent, and ex-gasoline dispersion was 2.2 percentage points. The median selected-essential inflation rate was 6.4 percent. This means the equal-weighted mean and standard-deviation dispersion measure are useful stress screens, but fragile if interpreted as representative household exposure.

That fragility is not a reason to discard the exercise. It is the central lesson: a credible household planning-risk index must distinguish energy-sensitive shocks from broader must-pay inflation, and must eventually use household-specific expenditure weights.

5. Unevenness, volatility, and category timing

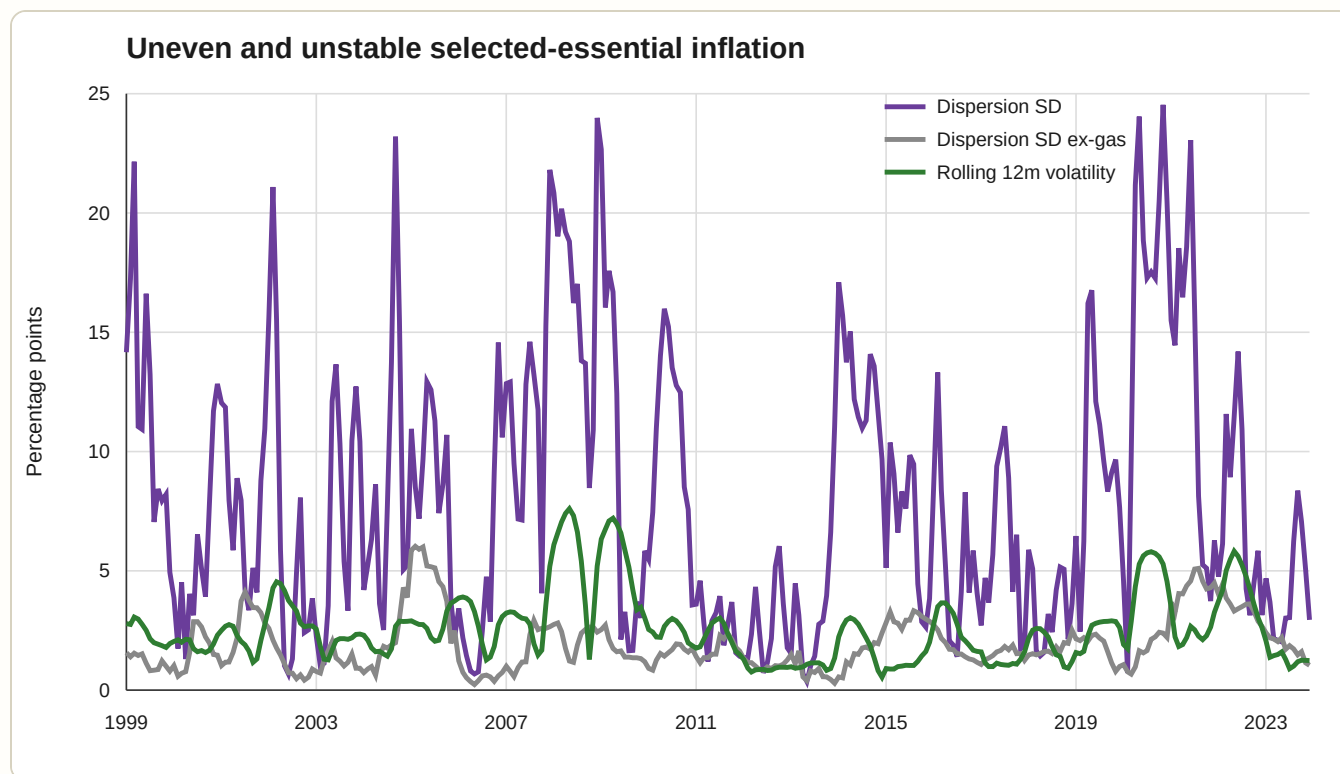


Figure 2. Selected-essential inflation dispersion, ex-gasoline dispersion, and rolling volatility.

Figure 2 separates three objects: cross-category dispersion, cross-category dispersion excluding gasoline, and rolling volatility in the selected-essential mean. Dispersion means selected categories are moving differently. Volatility means the measured basket changes unstably over time. Neither is the same as uncertainty, which would require comparing realized bills with expected bills.

Table 3. Component peak inflation timing during the recent shock.

Series	Peak YoY month	Peak YoY inflation
Headline CPI-U	2022-06	9.0
Food at home	2022-08	13.6
Rent of primary residence	2023-03	8.8
Electricity	2022-08	15.8
Gasoline	2022-06	59.8
Medical care	2022-09	6.0

Table 3 remains one of the more useful descriptive facts. Peaks in year-over-year inflation arrived at different times: headline CPI-U and gasoline in June 2022, food at home and electricity in August 2022, medical care in September 2022, and rent of primary residence in March 2023.

These are peaks in year-over-year inflation rates, not peaks in price levels. Rent also deserves special caution because CPI rent adjusts with leases and measurement lags; it is not a contemporaneous market-rent bill for every household.

6. Price levels, not only inflation rates

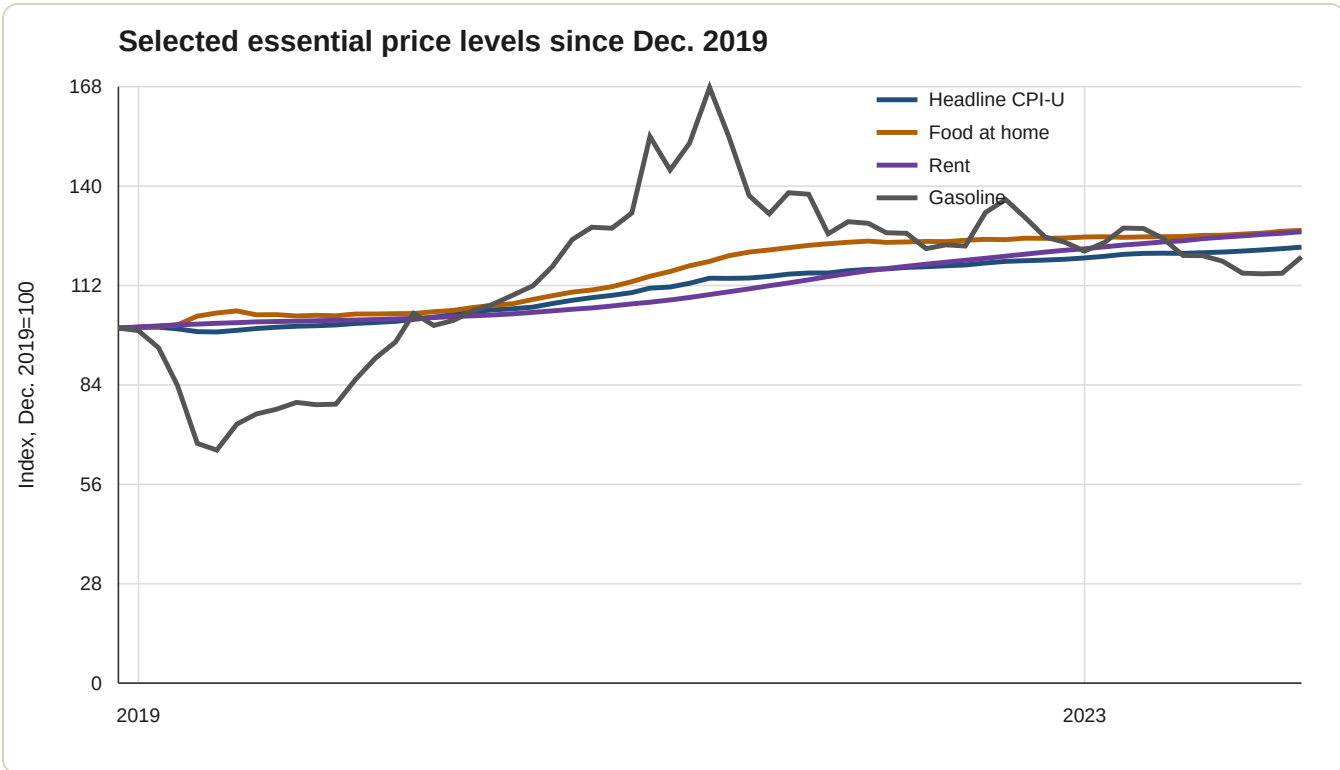


Figure 3. Selected essential price-level indexes since December 2019.

A household can feel pressure even after inflation rates fall because prices remain higher than before the shock. Figure 3 therefore adds price-level indexes normalized to December 2019.

Table 4. Price-level changes from December 2019 to December 2024.

Series	Dec. 2024 price index (Dec. 2019=100)	Cumulative change
Headline CPI-U	122.8	22.8
Food at home	127.5	27.5
Rent of primary residence	127.1	27.1
Electricity	131.9	31.9
Gasoline	120.0	20.0
Medical care	111.6	11.6

By December 2024, headline CPI-U was 22.8 percent above its December 2019 level. Food at home was 27.5 percent higher, rent of primary residence 27.1 percent higher, electricity 31.9 percent higher, gasoline 20.0 percent higher, and medical care 11.6 percent higher. These cumulative changes are

different from inflation-rate peaks. Both matter for household interpretation.

7. Sentiment as motivation only

The earlier version showed consumer sentiment beside dispersion. This revision removes that figure from the main evidentiary arc. Consumer sentiment is affected by employment, politics, asset prices, real wages, pandemic conditions, rates, debt burdens, and many other forces. Without controls or household-level exposure, a visual comparison risks overinterpretation.

Sentiment remains in Table 1 only as background context. The note does not claim that selected-essential dispersion caused sentiment to fall.

8. Planner relevance

A household-resilience planner should not use headline CPI alone when explaining budget instability. It should ask:

1. Which must-pay categories are rising fastest?
2. Are shocks concentrated in energy or broader across essentials?
3. Are price levels still elevated even after inflation rates cool?
4. Does the household have exposure to the volatile categories?
5. Does the household have enough liquidity to absorb category-specific surprises without missed payments or interrupted long-horizon plans?

The practical implication is not a universal formula. It is a measurement discipline: separate average inflation, selected-category inflation, cumulative price levels, dispersion, volatility, exposure, and outcomes.

9. Limits

This is a descriptive public-data companion note. It does not estimate household-specific inflation exposure. It does not use Consumer Expenditure Survey weights, SHED microdata, administrative transaction data, or household bills. It does not observe income timing, debt service, missed payments, delayed care, borrowing, or savings depletion. The selected-essential basket excludes many important household obligations, including natural gas, water and sewer, auto insurance, health insurance, child care, telecommunications, debt service, and mortgage obligations.

The rent category applies differently to renters and homeowners. Gasoline matters differently for commuters and non-drivers. Medical care matters differently by health status and insurance design. These differences are exactly why the current aggregate exercise should be treated as a measurement note rather than a household-stress estimate.

10. Next empirical step

A serious empirical paper would construct household-specific exposure measures such as $\pi_{it_H} = \sum_c s_{ic0} \pi_{ict}$, where s_{ic0} are pre-period household expenditure shares and π_{ict} are category inflation rates. The paper would then link exposure to outcomes such as missed payments, high-cost borrowing, food insecurity, delayed medical care, credit-card balance carrying, savings depletion, reduced retirement contributions, or subjective financial stress.

A plausible design would compare high- and low-exposure households during common category shocks, using a specification such as $Y_{it} = \alpha_i + \lambda_{t} + \beta(\text{Exposure}_{i0} \times \text{Shock}_t) + X_{it} \gamma + \epsilon_{it}$.

The key requirements would be pre-period exposure definitions, household or group fixed effects where possible, trend checks, and robustness to energy-driven shocks.

11. Conclusion

Headline CPI matters, but it is not the whole household planning-risk problem. Households face categories, bills, timing, liquidity constraints, and exposure differences. Selected essential prices during the recent inflation episode moved unevenly and at staggered times, while cumulative price levels remained elevated even after inflation rates cooled.

The revised claim is modest. Public aggregate data can motivate a better measurement framework, but they cannot measure planning damage by themselves. The right next step is household-level exposure linked to hardship outcomes. Until then, this note is best read as a transparent public-data companion: useful for explaining why aggregate CPI can miss household budget instability, but not a substitute for causal empirical work.

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